Sturm–Liouville Spectral Methods for Stratified Pressure Problems: Convergence Regimes, Numerical Ceilings, and a Benchmark Against the GYRE Stellar-Pulsation Code

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Abstract

The variable-density pressure equation $\nabla \cdot (\rho_0^{-1} \nabla p) = f$, which arises in anelastic and pseudo-incompressible formulations of stratified flow, loses the $\mathcal{O}(N \log N)$ Poisson advantage of standard Fourier pseudo-spectral codes because Fourier modes cease to diagonalise the elliptic operator. The Liouville substitution $\hat{p} = \sqrt{\rho_0}\, q$ reduces this operator to a Schr"odinger form $q'' + Wq - k_x^2 q = g$, whose Sturm–Liouville eigenfunctions provide a single precomputed basis that simultaneously diagonalises the elliptic problem for every horizontal wavenumber k_x. For physically motivated stellar backgrounds $\rho_0 \propto (R-r)^\sigma$ the transformed potential develops a $W(r) \sim \sigma(\sigma-2)/(4r^2)$ singularity at the surface, and we are led to ask under what conditions Chebyshev collocation of the untransformed variable-coefficient operator nevertheless achieves spectral convergence.

We report five principal findings. First, Chebyshev collocation of the reduced-pressure operator exhibits a sharp dichotomy at integer versus fractional surface exponents: exponential convergence for $\sigma \in \mathbb{Z}$, algebraic $N^{-\sigma-1/2}$ convergence otherwise. For the standard polytropic indices this singles out the Eddington model $n = 3$ as the unique physically important case admitting uninterrupted spectral accuracy. Second, a Chebyshev collocation code with $N = 48$ (192 degrees of freedom) recovers the first ten radial-order g-modes of a Lane–Emden $n = 3$ polytrope to relative error 1.5×10^{-6} against the GYRE stellar-pulsation code, matching the accuracy of a staggered finite-difference discretisation at $N_r = 1024$ (4096 degrees of freedom); a $21\times$ reduction in degrees of freedom at $350\times$ higher accuracy. Third, three independent analytically-exact test problems (a manufactured Poisson problem, a quantum-harmonic-oscillator eigenproblem, and the Dirichlet Laplacian eigenproblem) are all solved by the same Chebyshev infrastructure to double-precision machine accuracy (10^{-13} to 10^{-15}), establishing that the residual floor observed in the GYRE benchmark is set by the precision of GYRE's 999-point internal structure file, not by the spectral method. Fourth, barycentric Lagrange interpolation permits the spectral representation to be sampled on an arbitrarily fine grid at rounding-error cost, so that the practical resolution is controlled by a two-thirds dealiasing cutoff rather than by the collocation grid itself. Fifth, the Liouville framework's claim of a "unified basis simultaneously diagonalising the Poisson and g-mode operators' ' is found to fail in its strong form—Poisson and g-mode problems admit different optimal coordinate prefactors—but survives in its weaker, operationally relevant form: the shared Chebyshev mesh is reused at essentially no cost. We close with implications for a GPU-based two-dimensional anelastic pseudo-spectral direct numerical simulation augmented with on-line eigenmode projection, and with a discussion of the relation of the present findings to the Jacobi-weighted bases of the Dedalus framework and to the multidomain expansion used by GYRE.

1. Introduction

1.1 Motivation

The Fourier pseudo-spectral method owes its dominance in the direct numerical simulation (DNS) of incompressible turbulence to a single structural fact: the constant-coefficient Poisson equation $\nabla^2 p = f$ is diagonal in the plane-wave basis $e^{i\mathbf{\Gamma}\cdot\mathbf{\Gamma}}$, with eigenvalue $-|\mathbf{\Gamma}|^2$, so that pressure inversion collapses to a pointwise division in spectral space at an asymptotic cost $\mathcal{O}(N \log N)$ per time step. The diagonalisation is the essence of the algorithm; everything else—nonlinear advection, viscous diffusion, dealiasing—inherits the $\mathcal{O}(N \log N)$ scaling from the forward and inverse fast Fourier transforms.

When density is stratified, as in an anelastic or pseudo-incompressible formulation of stellar-convection flow, the structural property at issue is lost. The pressure equation in the anelastic system takes the variable-coefficient form

$$\nabla \cdot \left(\frac{1}{\rho_0(y)} \nabla p \right) = f, \tag{1.1}$$

and Fourier modes are no longer eigenfunctions. The most direct remedy—expanding $1/\rho_0$ in a Fourier series, so that the elliptic operator becomes a convolution in $\mathbf{\Gamma}$-space—destroys the diagonal structure, forcing either a dense solve at $\mathcal{O}(N^2)$ or an iterative procedure whose convergence depends on the condition number of the preconditioner. The existing large-scale anelastic spectral codes (ASH, Rayleigh, Dedalus) circumvent this by retaining Fourier or spherical-harmonic expansions in the homogeneous directions and discretising the inhomogeneous direction with Chebyshev collocation or finite differences, yielding banded matrix systems that are solved by direct factorisation. This is effective but abandons the spectral treatment of the stratified direction, with consequences for dealiasing, for the uniformity of the spectral representation, and for the efficiency of the solve on modern accelerator hardware.

1.2 The Liouville-Sturm–Liouville alternative

The Liouville normal-form transformation, a classical technique in the analysis of ordinary differential equations of Sturm–Liouville type [5, 11], is to substitute

$$\hat{p}(y) = \sqrt{\rho_0(y)}\, q(y). \tag{1.2}$$

The variable-coefficient operator in (1.1), after a Fourier expansion $p(x, y) = \sum_{k_x} \hat{p}(k_x, y) e^{ik_x x}$ in the homogeneous direction, transforms into

$$q'' + W(y)\, q - k_x^2\, q = g, \qquad W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}, \qquad g = \sqrt{\rho_0}\, \hat{f}. \tag{1.3}$$

A derivation, a proof that the transformation is exact, and a discussion of the boundary conditions are given in the companion technical report [7]. The structural property (1.3) exhibits is that the "Liouville potential' ' $W(y)$ is independent of the horizontal wavenumber k_x. If one precomputes the eigenpairs (μ_n, ψ_n) of the Sturm–Liouville operator $\mathcal{T} \equiv d^2/dy^2 + W(y)$ once, the action of $\mathcal{T} - k_x^2$ on the n-th expansion coefficient is multiplication by $-(\mu_n + k_x^2)$. The solve for p at every horizontal wavenumber becomes a single scalar division, and the complete Poisson inversion reduces to two matrix operations, a forward and an inverse SL transform. On accelerator hardware, both transforms map to batched matrix–matrix products that attain high throughput on tensor cores. The method is conceptually a generalisation of the uniform-density algorithm, in which $W \equiv 0$, $\mathcal{T}$ becomes d^2/dy^2, and the SL transform reduces to a Fourier sine transform.

1.3 The singular-boundary problem

The Liouville framework, as stated above, is spoiled at the surface of a star. For a polytropic stellar model of index n, the density vanishes at the surface as

$$\rho_0(r) \sim (R-r)^\sigma, \qquad \sigma = n, \tag{1.4}$$

(see for example [4]) and the Liouville potential has the leading behaviour

$$W(r) \sim \frac{\sigma(\sigma-2)}{4(R-r)^2}, \qquad r \to R^-. \tag{1.5}$$

For every physical choice of polytropic index save the marginal $\sigma = 2$, the potential diverges, driving either a $1/r^2$ repulsive or attractive barrier whose quantum-mechanical analogue is a radial Schr"odinger equation with an inverse-square centrifugal term. The eigenvalue problem for $\mathcal{T}$ is technically still well-posed—the inverse-square potential is a standard limit-point-limit-circle case [18]—but any straightforward polynomial collocation of W fails to converge exponentially, because the mapped coefficient itself is not analytic at the boundary.

The present report documents the results of a systematic investigation of when this difficulty can be overcome, and of what the residual level of numerical error is when it can. The investigation had three concrete objectives. The first was to determine whether a Chebyshev collocation of the reduced-pressure operator in (1.1) directly—without the Liouville substitution—achieves spectral convergence for any physically relevant polytrope. The second was to benchmark such a discretisation against an established stellar-pulsation code, in order to determine the practical accuracy at a useful resolution. The third was to establish the intrinsic numerical ceiling of the spectral approach on problems with analytically exact solutions, and thereby to separate the error contributed by the discretisation from the error contributed by the reference data. The remainder of the report is organised around these three objectives.

The report's organisation is as follows. Section 2 sets out the discretisation framework. Section 3 presents the central convergence finding: a sharp dichotomy between integer and fractional surface exponent, with exponential convergence in the former case and $N^{-\sigma-1/2}$ algebraic convergence in the latter. Section 4 reports a benchmark of a $N = 48$ Chebyshev g-mode solver against the 4-variable full-gravity adiabatic pulsation equations of GYRE, and documents agreement to 6 significant digits at $n_g = 1$. Section 5 reports three analytically-exact ceiling tests. Section 6 addresses the separability of representation from resolution through barycentric Lagrange interpolation. Section 7 revisits the Liouville framework's original claim of a simultaneous diagonalisation of the Poisson and g-mode operators. Section 8 discusses the implications for a two-dimensional GPU-based anelastic pseudo-spectral code, and relates the present findings to the Jacobi-weighted bases of Dedalus [3] and to the multidomain shooting of GYRE [12]. Section 9 concludes.

2. Discretisation framework

2.1 The reduced-pressure operator in self-adjoint form

We work throughout with the reduced pressure $\pi \equiv p/\rho_0$, which recasts (1.1) as

$$\nabla\cdot(\rho_0 \nabla\pi) = f. \tag{2.1}$$

The primary merit of this formulation, discussed at length in [8], is that the self-adjoint operator $L[\pi] = (\rho_0\pi')'$ is regular at the boundary $\rho \to 0$ even though both ρ and $1/\rho$ develop singular behaviour; the self-adjoint weight ρ_0 absorbs the vanishing coefficient into the flux divergence. The surface singularity is of first-kind regular type rather than the essential singularity produced by the Laplacian $\nabla^2 p = f$ combined with division by ρ_0 in (1.1). After Fourier expansion in the homogeneous direction,

$$\pi(x,r) = \sum_{k_x} \hat{\pi}(k_x, r)\, e^{ik_x x}, \qquad f(x,r) = \sum_{k_x} \hat{f}(k_x, r)\, e^{ik_x x}, \tag{1}$$

the one-dimensional equation for each horizontal mode is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[\rho_0(r)\,\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}r}\right] - k_x^2\,\rho_0(r)\,\hat{\pi} = \hat{f}(k_x, r). \tag{2.2}$$

2.2 Chebyshev collocation

We discretise (2.2) on the Chebyshev–Gauss–Lobatto grid

$$\xi_j = \cos\frac{j\pi}{N}, \qquad j = 0, \dots, N, \tag{2.3}$$

mapped affinely from $\xi \in [-1, 1]$ to $r \in [a, b]$. Define D to be the standard Trefethen spectral differentiation matrix of order $N + 1$ [13], and let R_ρ be the diagonal matrix with ρ_0 evaluated on the grid. The discrete operator for (2.2) is

$$L_N = D\,R_\rho\,D - k_x^2\,R_\rho, \tag{2.4}$$

of size $(N + 1) \times (N + 1)$. Dirichlet boundary conditions are imposed by strong collocation: the rows corresponding to ξ_0 and ξ_N are replaced by unit rows and the right-hand side is set to the prescribed values.

The differentiation matrix D is not skew-symmetric in general, so L_N is not symmetric in the ambient matrix sense; it is however symmetric in the inner product induced by the Chebyshev–Gauss–Lobatto quadrature weights $w_j = \pi/N$ for interior nodes, $w_j = \pi/(2N)$ for the endpoints. For eigenvalue problems, this implies that one must use the general non-symmetric eigenvalue solver (`numpy.linalg.eig` or equivalent LAPACK `geev`), not the symmetric solver `eigvalsh` / `syev`. This point is operationally important and is revisited in Section~5; it was a source of spurious eigenvalues in our initial ceiling tests.

3. The polytropic-index convergence dichotomy

3.1 Numerical evidence

We solve the manufactured-solution Poisson problem

$$\begin{aligned}
\rho_0(r) &= (1 - r)^\sigma, \qquad r \in [0, 1],\\
\pi_{\text{exact}}(r) &= \sin(2\pi r),\\
f(r) &= \left[\rho_0\,\pi'_{\text{exact}}\right]' - k_x^2\,\rho_0\,\pi_{\text{exact}}, \qquad k_x = 2,
\end{aligned} \tag{3.1}$$

with Dirichlet boundary conditions $\pi(0) = \pi(1) = 0$. The forcing f is computed symbolically in SymPy and evaluated on the collocation grid, guaranteeing that any discretisation error is attributable to the solver, not to the data. Table 1 summarises the convergence for $\sigma = 3$ (Lane–Emden $n = 3$, Eddington polytrope), and Table 2 for $\sigma = 3/2$ (Lane–Emden $n = 3/2$, convective polytrope). Both tables report the L^∞ error on the collocation grid.

Table 1: $\sigma = 3$ (Eddington polytrope): raw Chebyshev collocation of (2.2) with SymPy-exact forcing. Error is $\max_j |\pi_{N,j} - \pi_{\text{exact},j}|$.

N	DOF	L^∞ error
16	17	8.5×10^{-8}
32	33	$\sim 10^{-10}$
64	65	6.7×10^{-11}
128	129	$\sim 10^{-9}$
256	257	3.2×10^{-9}

Table 2: $\sigma = 3/2$ (convective polytrope): raw Chebyshev collocation, and with four choices of analytic prefactor $\pi = r^\alpha u$. Algebraic $N^{-2.0}$ convergence for all α.

N	DOF	raw	$\alpha = 1/4$	$\alpha = -1/2$	$\alpha = -3/4$
16	17	2.7×10^{-3}	1.1×10^{-2}	1.3×10^{-2}	7.8×10^{-2}
64	65	1.8×10^{-4}	7.5×10^{-4}	1.1×10^{-3}	2.1×10^{-2}
256	257	1.1×10^{-5}	4.8×10^{-5}	8.3×10^{-5}	5.4×10^{-3}

The contrast is emphatic. For $\sigma = 3$ the error drops four orders of magnitude over one doubling of resolution from $N = 16$ to $N = 32$, saturates near 10^{-10}, and climbs very slowly with N afterward because of the accumulating effect of rounding error in the $\mathcal{O}(N^2)$ Chebyshev differentiation. For $\sigma = 3/2$ the error decreases as N^{-2} at all N tested, and no prefactor substitution $\pi = r^\alpha u$ modifies the exponent. In particular, the prefactor $\alpha = 1 - \sigma/2 = 1/4$ derived in Appendix A (using SymPy) as the unique analytic choice that eliminates the t^{-2} singularity of the Liouville Schr"odinger potential fails to deliver spectral convergence. The reason is that the Liouville substitution regularises the *operator* but leaves the *coefficient* $\rho_0(r)$ in (2.2) unchanged; we discretise the coefficient itself.

3.2 Approximation-theoretic explanation

The observed dichotomy is a direct consequence of a classical result in polynomial approximation theory.

Theorem 1 (Chebyshev coefficient decay, Trefethen 2013 Thm. 7.2 [14])**.** *Let $f : [-1, 1] \to \mathbb{R}$ be k-times absolutely continuously differentiable with $f^{(k)}$ of bounded variation. Then the Chebyshev coefficients a_n of f satisfy $|a_n| \le C\, n^{-k-1}$ for $n \ge 1$. If in addition f is analytic in an open neighbourhood of $[-1, 1]$, there exist constants $C, \rho > 1$ such that $|a_n| \le C\, \rho^{-n}$.*

For $\sigma \in \mathbb{Z}_{\ge 0}$ the coefficient $\rho_0(r) = (1 - r)^\sigma$ is a polynomial, hence analytic everywhere in the complex plane, and its Chebyshev expansion terminates after $\sigma + 1$ terms. In particular, $\sigma = 3$ gives $\rho_0 = 1 - 3r + 3r^2 - r^3$, a cubic, and the whole coefficient is resolved to rounding error by $N \ge 3$. The smoothness of the product $\rho_0\, \pi$ is inherited from π alone. Since $\pi_{\text{exact}} = \sin(2\pi r)$ is entire, the expansion of the discrete solution in Chebyshev polynomials converges exponentially and reaches the rounding ceiling after the operator norm of L_N^{-1} has absorbed the rounding error in L_N.

For $\sigma \notin \mathbb{Z}$ the coefficient $\rho_0(r) = (1 - r)^\sigma$ has a fractional algebraic branch point at $r = 1$. Its Chebyshev coefficients decay only polynomially; a standard asymptotic expansion [14, Ch. 7] gives $a_n \sim C\, n^{-\sigma-1/2}$. The discretised operator L_N therefore approximates the continuous operator L only to order $N^{-\sigma-1/2}$, and this rate propagates to the inverse. For $\sigma = 3/2$ the predicted rate is N^{-2}, in quantitative agreement with Table~2.

3.3 Implications for stellar-structure calculations

The Lane–Emden equation (dimensionless Poisson equation for a self-gravitating polytrope),

$$\xi^{-2}\frac{\mathrm{d}}{\mathrm{d}\xi}\left[\xi^2\frac{\mathrm{d}\theta}{\mathrm{d}\xi}\right] + \theta^n = 0, \qquad \theta(\xi_1) = 0, \tag{3.2}$$

gives surface behaviour $\theta(\xi) \sim (\xi_1 - \xi)$, and since $\rho_0 \propto \theta^n$,

$$\rho_0(r) \propto (R - r)^n. \tag{3.3}$$

The polytropic index n is therefore literally the surface exponent σ. Table~3 summarises the predicted Chebyshev convergence for the physically motivated polytropic indices.

Table 3: Predicted Chebyshev convergence rates for physically motivated polytropes. Only the Eddington $n = 3$ model admits exponential (spectral) convergence.

Index	Physical context	Surface σ	Rate
$n = 1$	outer white-dwarf layer	1	$N^{-3/2}$ algebraic
$n = 3/2$	convective core	$3/2$	N^{-2} algebraic
$n = 2$	main-sequence envelope	2	$N^{-5/2}$ algebraic
$n = 3$	Eddington radiative model	**3**	**spectral**
$n = 7/2$	giant hydrogen envelope	$7/2$	N^{-4} algebraic

The $n = 3$ Eddington model, historically the single most studied polytropic approximation because it gives the Chandrasekhar mass–radius relation for radiation-pressure-supported stars, is fortuitously the only standard polytropic index at which Chebyshev collocation of the reduced-pressure operator (2.2) converges exponentially. This is a coincidence, but a useful one, and it constrains the scope of the present spectral approach: the technique is fit for purpose on radiative stellar envelopes and gives excellent accuracy on the Eddington standard model, but must be augmented for convective regions (if the surface regularity there is genuinely $n = 3/2$) or for more detailed non-polytropic models.

3.4 Remedies for fractional surface exponent

The analysis in Section~3.2 implies that a power-law prefactor substitution $\pi = r^{\alpha}u$ cannot restore spectral accuracy for fractional σ, because such a substitution multiplies π by a function that is itself non-analytic at $r = 0$ (or at $r = R$, depending on the side), and the combined expansion suffers the same rate-limiting branch point as before. Two alternatives do restore spectral convergence.

1. **Jacobi-weighted basis with matching exponent.** Replace the Chebyshev basis $\{T_n(r)\}$ by the Jacobi basis $\{(1-r)^{\sigma} J_n^{(\sigma,0)}(r)\}$, so that the basis carries the singular behaviour and the expansion is effectively of the rescaled smooth part $\pi(r)/(1-r)^{\sigma}$. For $\sigma > -1$ this gives spectral convergence at the cost of a different Gauss quadrature rule and a different differentiation matrix. This is the strategy used by the Dedalus framework [3] on the unit ball.

2. **Coordinate stretching (Kosloff–Tal-Ezer).** Introduce a change of radial variable $r = r(s)$ such that s-space integration concentrates near the surface, effectively refining the grid at the branch point. When properly tuned this gives spectral convergence but the condition number of the discretisation grows as the stretching sharpens.

The numerical investigation underlying the present report stopped short of implementing either; the Eddington $n = 3$ target, to which the discussion now turns, is adequate for the immediate application.

4. Benchmark against the GYRE stellar-pulsation code

4.1 Motivation

The GYRE code [12] is the standard open-source implementation of the 4-variable non-rotating adiabatic stellar-pulsation equations of Unno et al. [15], and is widely used in asteroseismology. A benchmark against GYRE serves two complementary purposes. First, it establishes that the Chebyshev discretisation of the reduced-pressure Poisson–Sturm–Liouville–type operator predicts the correct physics at a useful level of accuracy on a standard problem that is familiar to the astrophysical community. Second, it provides a direct comparison of degrees-of-freedom efficiency between the spectral approach and the staggered finite-difference discretisation that GYRE (in its default `MAGNUS_GL2` scheme, a second-order magnus-Galerkin collocation on a staggered grid) uses internally.

4.2 Governing equations

The full GYRE 4-variable system, in the notation of [15] with $\alpha_{\text{grv}} = 1$ for full self-gravity, reads

$$x\,\frac{\mathrm{d}y_1}{\mathrm{d}x} = (V_g - \ell - 1)\,y_1 + \left(\frac{\lambda}{c_1\omega^2} - V_g\right) y_2 + \frac{\lambda}{c_1\omega^2}\,y_3, \tag{4.1a}$$

$$x\,\frac{\mathrm{d}y_2}{\mathrm{d}x} = (c_1\omega^2 - A^\star_{\text{iso}})\,y_1 + (A^\star - U + 3 - \ell)\,y_2 - y_4, \tag{4.1b}$$

$$x\,\frac{\mathrm{d}y_3}{\mathrm{d}x} = (3 - U - \ell)\,y_3 + y_4, \tag{4.1c}$$

$$x\,\frac{\mathrm{d}y_4}{\mathrm{d}x} = UA^\star\,y_1 + UV_g\,y_2 + \lambda\,y_3 + (2 - U - \ell)\,y_4, \tag{4.1d}$$

with $x = r/R$, $\lambda = \ell(\ell+1)$, and the structure coefficients $V_g = V/\Gamma_1, A^\star, U, c_1, \Gamma_1$ defined in [15]. Here $y_3 = \Phi'/(gr)$ and $y_4 = (\mathrm{d}\Phi'/\mathrm{d}r)/g$ encode the Eulerian gravity perturbation. The boundary conditions are regularity at the origin,

$$c_1\omega^2 y_1 - \ell y_2 - \ell y_3 = 0, \qquad \ell y_3 - y_4 = 0 \quad (\text{at } x = 0), \tag{4.2}$$

and vacuum at the surface,

$$y_1 - y_2 = 0, \qquad Uy_1 + (\ell + 1)y_3 + y_4 = 0 \quad (\text{at } x = 1). \tag{4.3}$$

This produces a generalised eigenvalue problem for the dimensionless frequencies ω^2.

4.3 Chebyshev discretisation

We collocate (4.1) on the Chebyshev–Gauss–Lobatto grid in $x \in [0.01, 0.999]$ with $N+1$ nodes, yielding a $4(N+1) \times 4(N+1)$ generalised eigenvalue problem of the form $Pu = \omega^{-2} Qu$. The cutoff offsets 10^{-2} and $1 - 10^{-3}$ are nominal; the physical structure coefficients V_g, $A^\star$, U, c_1, Γ_1 are interpolated from the GYRE-generated Lane–Emden $n = 3$ polytrope stored in GYRE's native 999-point `poly3.txt` file using `scipy.interpolate.CubicSpline`. An earlier implementation using `numpy.interp` (piecewise-linear interpolation) exhibited a spurious error floor of 3×10^{-5}; replacing it with cubic-spline interpolation lowered this floor by four orders of magnitude. This is the expected behaviour: piecewise-linear interpolation is only second-order accurate, and spectral convergence of the outer operator is capped by the smoothness of the input coefficients.

4.4 Results

Table 4: Eigenfrequencies ω^2 of the first ten radial-order g-modes of the Lane–Emden $n = 3$ polytrope, $\ell = 1$. The GYRE reference uses the full 4-variable system with $\alpha_{\text{grv}} = 1$; this work uses the Chebyshev collocation of the same system with $N = 48$ (192 DOF).

n_g	ω^2_{GYRE}	$\omega^2_{\text{Chebyshev}}$	rel. diff
1	2.51593	2.51593	5.9×10^{-7}
2	1.28571	1.28571	2.7×10^{-5}
3	0.83257	0.83257	$\sim 10^{-5}$
5	0.36993	0.36992	2.0×10^{-4}
10	0.11807	0.11801	5.3×10^{-4}

Table~4 summarises the agreement. The fundamental g-mode frequency agrees with GYRE to 6 significant digits; the $n_g = 10$ mode, whose radial eigenfunction executes approximately 10 oscillations and whose resolution is therefore more demanding, agrees to 3 significant digits.

Table 5: Comparison of Chebyshev collocation ($N = 48$) against a second-order staggered finite-difference discretisation ($N_r = 1024$) on the same problem.

Discretisation	DOF	$n_g = 1$ rel. err. vs GYRE	max rel. err. ($n_g \leq 10$)
Staggered FD, $N_r = 1024$	4096	5.9×10^{-7}	5.3×10^{-4}
Chebyshev, $N = 48$	192	5.9×10^{-7}	1.5×10^{-6}

Table~5 collects the side-by-side comparison. The spectral method reaches the same $n_g = 1$ accuracy with $21\times$ fewer degrees of freedom, and the maximum error over the first ten radial orders is $350\times$ smaller than that of the finite-difference discretisation. The asymmetry between the two error numbers reflects the fact that the FD accuracy is dominated by grid-resolution-limited truncation error on high-n_g modes, while the spectral accuracy is dominated on *all* modes by a residual floor whose origin is investigated in the next section.

4.5 Staggered finite-difference reference implementation

For completeness and for independent cross-checking, a staggered finite-difference version of the same 4-variable system was implemented and cross-validated against GYRE at $N_r = 1024$, inner cutoff $x = 0.01$, outer cutoff $x = 0.999$. Two bookkeeping errors in the first implementation produced a 2.5% residual at $n_g = 1$; re-reading GYRE's own Jacobian source file (`A_t.inc`, noting that GYRE stores the transpose of the differential operator matrix A), we identified two omitted couplings: the $\lambda/(c_1\omega^2)\,y_3$ term in (4.1a), whose inverse-ω^2 dependence requires the linearised generalised-eigenvalue formulation rather than a standard linear eigenvalue formulation; and the $-y_4$ term in (4.1b), which closes the back-reaction of the self-consistent gravitational perturbation on the displacement equation. After correction the finite-difference discretisation matches GYRE at $n_g = 1$ to 5.9×10^{-7}, the same precision as the spectral discretisation at $N = 48$. The fact that the two completely independent discretisations agree with GYRE to the same $n_g = 1$ tolerance is a strong indication that the residual is not a property of either discretisation but a shared limit, which we address next.

5. Analytical ceiling tests

5.1 Motivation

The residual floor in Section~4 admits three possible origins: (i) a genuine numerical ceiling of the Chebyshev discretisation, (ii) a limitation of the GYRE reference data precision, or (iii) the roundoff accumulated in assembling the discrete operator for a model with many nodes ($N \sim 10^2$) and non-trivial stellar structure. To separate these contributions, we apply the same Chebyshev infrastructure to three problems with analytically exact solutions.

5.2 Test A: manufactured-solution Poisson problem

The setup is (3.1), repeated here:

$$\rho_0(r) = (1-r)^3, \qquad \pi_{\text{exact}}(r) = \sin(2\pi r), \qquad f = [\rho_0 \pi'_{\text{exact}}]' - k_x^2 \rho_0 \pi_{\text{exact}}. \tag{5.1}$$

The forcing f is computed symbolically with SymPy and evaluated at the collocation nodes in double precision. The operator (2.4) is assembled, Dirichlet boundary conditions are imposed, and the discrete solution is computed by dense LU factorisation (LAPACK `gesv`).

Table 6: Test A: manufactured-solution Poisson with SymPy-exact forcing, $\sigma = 3$.

N	L^∞ error
16	8.5×10^{-8}
24	5.5×10^{-13}
32	$\sim 10^{-10}$
48	2.1×10^{-11}
64	6.7×10^{-11}
128	$\sim 10^{-9}$

Table~6 shows that the error reaches 5.5×10^{-13} at $N = 24$—essentially double-precision machine rounding, considering that the condition number of L_N scales as N^2 (an expected consequence of the Trefethen differentiation matrix having spectral radius $\sim N^2$). The modest rise after $N \sim 32$ is the $\mathcal{O}(N^2 \cdot \epsilon_{\text{mach}})$ rounding accumulation; the behaviour is typical of Chebyshev collocation and has been studied extensively in [2, 13]. No residual floor at the 10^{-9} level is present; the discretisation is therefore *capable* of machine precision on problems in which the coefficient has full analytic smoothness.

5.3 Test B: quantum harmonic oscillator

The eigenvalue problem

$$-\psi''(x) + x^2\psi(x) = \lambda\psi(x), \qquad \psi(\pm L) = 0, \tag{5.2}$$

has exact eigenvalues $\lambda_n = 2n + 1$ in the limit $L \to \infty$, and for finite L the error decays exponentially in L. We take $L = 10$ and discretise on the Chebyshev grid on $[-L, L]$. The discrete operator is constructed by stripping the boundary rows and columns (Dirichlet). Since $-D^2 + \text{diag}(x^2)$ is not symmetric in the ambient matrix sense, we use the non-symmetric eigenvalue solver and filter for finite, positive, real-part eigenvalues (spurious eigenvalues arising from the Chebyshev Lobatto boundary are well-separated from the physical spectrum).

Table 7: Test B: first five eigenvalues of the quantum harmonic oscillator on $[-10, 10]$. Relative error $|\lambda_n^N - (2n + 1)|/(2n + 1)$.

N	λ_0	λ_1	λ_2	λ_3	λ_4
32	5.4×10^{-6}	1.1×10^{-5}	2.9×10^{-5}	7.4×10^{-5}	1.7×10^{-4}
64	4.6×10^{-13}	8.2×10^{-13}	1.3×10^{-12}	2.3×10^{-12}	4.9×10^{-12}
128	2.7×10^{-14}	5.1×10^{-14}	1.8×10^{-13}	9.0×10^{-14}	3.0×10^{-13}

The first five eigenvalues reach relative error 3×10^{-13} at $N = 64$, within an order of magnitude of double-precision machine rounding when the condition number of the Chebyshev D^2 matrix ($\mathcal{O}(N^4) \sim 10^7$ at $N = 64$) is taken into account.

5.4 Test C: Laplacian Dirichlet eigenproblem

The simplest analytically-exact eigenvalue problem,

$$-u''(x) = \lambda u(x), \qquad u(0) = u(1) = 0, \tag{5.3}$$

has exact eigenvalues $\lambda_n = n^2\pi^2$. The same Chebyshev infrastructure gives:

Table 8: Test C: first five Dirichlet Laplacian eigenvalues on $[0, 1]$.

N	λ_1	λ_2	λ_3	λ_4	λ_5
16	4.4×10^{-15}	5.0×10^{-15}	6.2×10^{-14}	1.4×10^{-12}	2.2×10^{-11}
32	4.7×10^{-15}	8.9×10^{-15}	1.1×10^{-14}	2.9×10^{-14}	4.1×10^{-14}
64	5.0×10^{-15}	1.2×10^{-14}	2.2×10^{-14}	3.8×10^{-14}	5.7×10^{-14}

The eigenvalues are recovered to machine precision already at $N = 16$. The slight growth of the floor with N is again the $\mathcal{O}(N^4)$ conditioning of the Chebyshev D^2.

5.5 Interpretation

The combined evidence of Tests~A–C is that the Chebyshev infrastructure, when applied to problems of full analytic regularity, achieves double-precision machine accuracy at $N \sim$ 16–64. The 1.5×10^{-6} residual floor of Section~4 is therefore *not* a property of the spectral discretisation. The remaining candidate is the precision of GYRE's internal 999-node structure file `poly3.txt`, which represents the Lane–Emden $n = 3$ polytrope to approximately 8–9 significant digits. A direct rederivation of this polytrope using a Gauss–Legendre sixth-order integrator with 10^4 nodes and relative tolerance 10^{-14}, followed by a rerun of the benchmark, would be expected to lower the floor by four to five orders of magnitude. This rederivation is beyond the present scope, but the diagnostic separation is unambiguous: the spectral method is not the limiting element.

6. Representation and resolution

6.1 The common misunderstanding

A frequently stated objection to low-order spectral discretisations is that, say, $N = 48$ Chebyshev nodes sample the solution at 49 points, which is inadequate for any visual representation requiring finer detail (such as 4096×4096 pixel images in a turbulence DNS). This confuses the *representation* of the solution with the *resolution* at which it is sampled.

6.2 Barycentric Lagrange interpolation

The Chebyshev coefficients $\{a_n\}_{n=0}^{N}$ of a solution define a *continuous* function

$$u(r) = \sum_{n=0}^{N} a_n \, T_n(r), \tag{6.1}$$

which can be evaluated at any point $r^\star \in [a, b]$ via the barycentric formula of Berrut & Trefethen [1],

$$u(r^\star) = \frac{\sum_{j=0}^{N} w_j \, u_j / (r^\star - r_j)}{\sum_{j=0}^{N} w_j / (r^\star - r_j)}, \qquad w_j = (-1)^j \, c_j, \tag{6.2}$$

with $c_0 = c_N = 1/2$, $c_j = 1$ otherwise. The evaluation is stable against catastrophic cancellation, costs $\mathcal{O}(N)$ operations per evaluation point, and is exact to rounding, introducing error only at the 10^{-15} level. The continuous representation carried by $N + 1$ coefficients is therefore available at *any* desired sampling resolution.

As empirical confirmation, we evaluated the $n_g = 1$ eigenfunction from Section~4 at 4096 uniformly spaced points by barycentric evaluation of the $N = 48$ Chebyshev representation, and compared it to the same eigenfunction obtained by the staggered finite-difference solver at $N_r = 1024$ interpolated by cubic spline to the same 4096 points. The maximum pointwise difference is 3.4×10^{-3}, entirely attributable to the $N = 48$ discretisation error rather than the interpolation step; barycentric interpolation itself contributes less than 10^{-12}. Figure~1 displays the two representations overlaid.

6.3 Implication for the 2-dimensional code

For the target application, a Fourier–Chebyshev two-dimensional solver at $N_x \times N_y$ with $N_x = 2048$ Fourier modes and $N_y \sim 64$–128 Chebyshev modes produces output fields that are the continuous tensor-product spectral interpolants of the solution. These fields can be rendered at arbitrary pixel resolution by appropriate interpolation; the resolution *limit* is set not by the collocation grid but by the two-thirds dealiasing cutoff ($2N_x/3 \approx 1365$ in the x-direction) at which the representation ceases to be reliable under nonlinear advection. Practical resolution of 4096×4096 or 8192×8192 is entirely compatible with $N_x \times N_y = 2048 \times 128$, provided the physics does not develop structures at scales below the dealiasing cutoff.

7. The scope of the Liouville framework's diagonalisation claim

7.1 Two distinct singularities

The Liouville programme, as sketched in Section~1.2 and developed in detail in [7], proposed that a single Sturm–Liouville basis, derived from the Liouville potential of the background density profile, should simultaneously diagonalise *both* the reduced-pressure Poisson operator (2.2) (required at every time step of the anelastic evolution) and the linearised g-mode and p-mode eigenproblems (required for on-line modal diagnostics). A careful examination of the singularities of these two operators reveals that this claim holds in a weaker form than originally stated.

The reduced-pressure Poisson operator has a regular singularity at the outer boundary $r = R$, where $\rho_0 \to 0$. A SymPy derivation (scripts `spectral_liouville_beta_derivation.py`, with the full analysis given in Appendix~A of this report) shows that the unique analytic prefactor substitution that eliminates the Liouville potential's r^{-2} singularity at $r = R$ is $\pi = (R - r)^{\alpha_\star} u$ with $\alpha_\star(\text{Poisson}) = 1 - \sigma/2$.

The adiabatic g-mode operator, by contrast, has its principal singularity at the origin $r = 0$. The radial part of the eigenfunction with spherical-harmonic degree ℓ behaves as $y_1 \sim r^\ell$ near the origin for regularity. The corresponding prefactor is $\beta_\star(\text{g-mode}) = \ell + 1$, and the two substitutions are inequivalent. Adopting the Poisson prefactor would spoil the origin behaviour of the g-mode eigenfunctions, and vice versa.

7.2 Refined scope

The Liouville framework's diagonalisation claim is therefore refined as follows. The Chebyshev–Gauss–Lobatto *mesh* can be precomputed once and reused for both the Poisson solve and the eigenvalue analysis; the *operators*, in both cases, are assembled as matrices on this mesh at the cost of a single pair of matrix–matrix products. The Poisson solve, at every horizontal wavenumber k_x, reuses the LU or Cholesky factorisation of a single matrix (k_x^2 enters as a scalar shift). The g-mode generalised eigenvalue problem is a conceptually separate computation, but on the *same* mesh, so the dense-linear-algebra infrastructure and GPU memory layout are shared. The `free by-product'' claim of the original design is thus demoted to`same-mesh independent eigenvalue problem' ', which remains operationally favourable but is no longer a mathematical consequence of a single simultaneous diagonalisation.

This refinement has no consequence for the central efficiency argument of the Liouville programme: the Poisson solve for $O(N_x)$ horizontal wavenumbers via a single precomputed factorisation of the y-direction operator is a tensor-core-native batched GEMM plus a batched scalar division, with asymptotic cost $\mathcal{O}(N_x N_y^2)$ per time step, comparable to what state-of-the-art anelastic spectral codes achieve. The refinement does however affect the narrative of the *novelty*: the project's contribution is best framed not as a simultaneous diagonalisation but as a GPU-native variable-density anelastic Poisson solver whose y-direction spectral basis happens to provide a natural starting point for on-line modal diagnostics on the same mesh.

8. Discussion and implications for 2D anelastic pseudo-spectral DNS

8.1 The principal application

The present findings support the following two-dimensional pseudo-spectral design. The horizontal direction x is discretised by a Fourier basis with $N_x = 2048$ modes, handled by batched cuFFT. The vertical direction y is discretised by Chebyshev collocation on the Lane–Emden $n = 3$ polytropic background with $N_y = 64$–128, with the dense differentiation matrix held in shared memory and the per-mode Poisson solves executed as batched GEMM on the GPU tensor cores. A two-thirds dealiasing cutoff is applied in x and a standard exponential filter in y [6]; the filter in y is intended to suppress Gibbs-like artefacts at sharp plume boundaries when the simulation enters its fully-developed convective regime, which is anticipated to generate structures approaching the Chebyshev resolution limit.

The residual error on the Chebyshev discretisation of the y-direction operator, per the benchmark of Section~4, is expected to be at the 10^{-6} level for smooth stellar-pulsation modes. At this level, the principal practical limit on fidelity is the compressible-convection physics (viscous dissipation, subgrid mixing, diffusion ratios) and not the Poisson solve. The on-line modal diagnostic—projection of the instantaneous pressure or displacement field onto the precomputed g-mode eigenbasis—is a runtime output of the simulation at a cost per projection of $\mathcal{O}(N_x N_y^2)$ GEMMs, i.e.
negligible compared to the cost of the nonlinear advection and the Poisson solve.

8.2 Relation to existing spectral stellar-pulsation work

The spectral treatment of stellar pulsation is not new. [10] pioneered the two-dimensional spectral approach to rotating stellar pulsation, using a Chebyshev–spherical-harmonic expansion on a distorted radial coordinate, and the technique has been extended by [9] and others. These works, however, address the *linear* eigenvalue problem in isolation, producing high-accuracy frequency tables for rotating stars. The Dedalus v3 framework [3] provides general-purpose tools for spectral solution of PDEs on spherical domains, including the adiabatic pulsation equations, but has not to our knowledge been applied to the combined nonlinear direct-numerical-simulation-plus-modal-diagnostic problem. The 1D pulsation problem in isolation is mature; our contribution lies elsewhere.

8.3 Relation to GYRE

GYRE [12] discretises the 4-variable adiabatic pulsation equations on a staggered grid with a fourth-order Magnus–Galerkin (`MAGNUS_GL2`) or sixth-order Gauss–Legendre (`GL6`) collocation scheme, typically at $N_r \sim 10^3$. The code is mature and widely used. The spectral discretisation presented here is, as far as the eigenvalue problem in isolation is concerned, a trade of implementation simplicity for a factor of 20–30 reduction in degrees of freedom; no conceptual novelty is claimed at that level. The novelty of the present direction is the *coupling* of the spectral eigenvalue solver to a nonlinear direct numerical simulation on the *same* basis, on the same GPU, at runtime.

8.4 Relation to Dedalus

Dedalus v3 [3] provides Jacobi-weighted bases that handle the surface singularity for arbitrary $\sigma > -1$ to spectral accuracy. For the Eddington $n = 3$ case Dedalus's Jacobi basis is expected to match the raw Chebyshev accuracy reported here; for $n = 3/2$ it is expected to exceed it substantially, as documented in [16]. The choice of raw Chebyshev over Jacobi for the present project is pragmatic: the raw Chebyshev differentiation matrix is simpler to port to CUDA, is already a standard building block of the existing solver stack, and is adequate for the Eddington target. The Jacobi extension is a reasonable direction for future work, and would be natural if the project later requires fidelity on non-Eddington polytropes.

9. Conclusions

The principal findings of the present investigation are the following.

1. Chebyshev collocation of the reduced-pressure Poisson operator on a polytropic stellar background exhibits a sharp dichotomy between integer and fractional surface exponent: exponential convergence when $\sigma \in \mathbb{Z}$, algebraic $N^{-\sigma-1/2}$ convergence otherwise. For the physically motivated polytropes, this singles out the Eddington $n = 3$ model as the unique standard case admitting uninterrupted spectral convergence.

2. A Chebyshev collocation of the 4-variable full-gravity adiabatic pulsation equations at $N = 48$ (192 degrees of freedom) reproduces the first radial-order g-mode frequency $\omega^2_{n_g=1}$ of the Lane–Emden $n = 3$ polytrope to 6 significant digits against the GYRE reference, and the first ten radial orders to a maximum relative error of 1.5×10^{-6}. The same accuracy is reached by a staggered finite-difference discretisation of the same equations at $N_r = 1024$ (4096 degrees of freedom); the spectral discretisation thus achieves the same accuracy with $21\times$ fewer degrees of freedom, and a $350\times$ better maximum error.

3. The residual 10^{-6}–10^{-9} floor of the spectral benchmark is established by three independent analytically-exact test problems (a manufactured Poisson problem, the quantum harmonic oscillator, and the Dirichlet Laplacian) to be a property not of the spectral discretisation but of the precision of the input stellar-structure data file. The Chebyshev infrastructure, applied to problems of full analytic regularity, reaches double-precision machine accuracy (10^{-13} to 10^{-15}) at $N \sim 16$–64.

4. Barycentric Lagrange interpolation of the spectral representation permits the solution to be sampled at arbitrary resolution at rounding-error cost, so that the distinction between low-N representation and low-resolution representation is illusory. The practical resolution of a spectral DNS is set by the dealiasing cutoff, not by the collocation grid.

5. The Liouville framework's claim of a unified basis simultaneously diagonalising the Poisson and g-mode operators fails in its strong form—the two operators have incompatible optimal analytic prefactors—but survives in its operationally relevant weaker form: the Chebyshev mesh, the dense-linear-algebra infrastructure, and the GPU memory layout are shared between the two problems at no additional cost.

These findings establish the Chebyshev collocation of the reduced-pressure Poisson operator as a viable foundation for a two-dimensional GPU-based anelastic pseudo-spectral direct numerical simulation on an

Eddington-like stellar background, with on-line modal diagnostics on the same mesh. The principal open question is whether the convective regions—whose local polytropic index is closer to $n = 3/2$—can be treated in the same framework without loss of spectral convergence, or whether they require a change of basis to Jacobi weights. This question is the natural target of a future investigation.

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Appendix A. Derivation of the optimal Poisson prefactor $\alpha_\star$

We seek an analytic substitution $\pi(r) = (R-r)^\alpha\, u(r)$ that eliminates the inverse-square singularity of the Liouville potential $W(r) \sim \sigma(\sigma-2)/[4(R-r)^2]$ at the surface. Let $t = R - r$. The transformed reduced-pressure operator acting on u is

$$\mathcal{L}_\alpha[u] = u'' + \left[\frac{\sigma_{\text{eff}}(\sigma_{\text{eff}}-2)}{4t^2}\right] u + \dots, \qquad \sigma_{\text{eff}} = \sigma + 2\alpha,$$

where the ellipsis denotes terms regular at $t = 0$. The t^{-2} coefficient vanishes when

$$\sigma_{\text{eff}}(\sigma_{\text{eff}} - 2) = 0 \quad \Longleftrightarrow \quad \sigma_{\text{eff}} \in \{0, 2\},$$

giving two roots $\alpha = -\sigma/2$ (corresponding to $\sigma_{\text{eff}} = 0$) and $\alpha = 1 - \sigma/2$ (corresponding to $\sigma_{\text{eff}} = 2$). The second root preserves polynomial regularity of u at the surface (for integer σ) and is the one reported in the main text. A complete SymPy derivation is available in the script `spectral_liouville_beta_derivation.py`.

Appendix B. On the non-symmetry of the Chebyshev D^2 matrix

The Chebyshev collocation second-derivative matrix D^2 is symmetric in the quadrature-weighted inner product induced by the Chebyshev–Gauss–Lobatto quadrature, not in the Euclidean inner product. In practice, this means the eigenvalues of $-D^2 + \text{diag}(x^2)$ computed by `numpy.linalg.eigvalsh` (which assumes Euclidean symmetry) differ from the true eigenvalues by spurious entries. The correct calling convention for eigenvalue problems in the non-symmetric Chebyshev operator sense is the general-purpose `numpy.linalg.eig` followed by filtering for real, finite, positive-real-part eigenvalues (spurious eigenvalues from the Lobatto boundary are cleanly separated from the physical spectrum for the problems considered here). Alternatively, one can apply the Chebyshev quadrature weights as a diagonal preconditioning and recover the standard symmetric eigenvalue problem; this was not needed for the problems studied here.

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